

ELWARD SYSTEMS -- OPERATIONAL CONTROLLED DOCUMENT

MECHANICAL ASSEMBLY DRAWINGS

JOB: 25036 | RELEASE: 1 | REV: A | DATE: 2026-08-24 | Parts Prep & 3-Row Assembly Workstations

1. Frame Assembly & Stiffener Placement

- * Panel Sub-Assembly Sequence: Perimeter Frame -> Structural Clips -> Intermediate Stiffener -> Gasket
- * Rivet Pattern: 3/16" 316 Stainless Steel blind rivets @ 12.0" maximum center-to-center.
- * Corner Joints: Stake corner keys with pneumatic crimper or fasten with #10-16 SS screws.
- * Structural Adhesive: Apply continuous structural silicone bead (Dow Corning 983 / Sika 205) along 4002 stiffeners.

2. Quality Checkpoint Checklist

- ☐ Squareness Tolerance: Diagonal check within +/- 1/16" across opposite corners.
- ☐ Flange Depth: Uniform 1.00" return without oil canning or waviness.
- ☐ Rivet Seating: Flush mandrel break with no protruding burrs or loose heads.
- ☐ Gasket Insertion: Fully seated continuous EPDM gasket without stretching or bunching.